

LAB 3 – WRITTEN JUSTIFICATION

System: Household Carbon Footprint & Target Tracker

Architecture: Layered Architecture

Name: Anusri Sharma

SRN: PES1UG25CS803

Household Carbon Footprint & Target Tracker

Architecture Selection:

We selected Layered Architecture as a design paradigm for our system. Our Household Carbon Footprint & Target Tracker system is composed of three logical layers, the Presentation Layer, the Business Layer and the Data Layer. The application layer design approach is to make the system clearer and easier to maintain by dividing the user interaction with the system, the application processing and data storage into different layers. The Lab 3 handout explains Layered Architecture as horizontal levels: Presentation, Business and Data layers.

Reason 1 - Separation of Responsibilities:

The additional Advantages of Layered Architecture include:- Clear separation of responsibilities between the components of the system. - User Interface Component (Resident and Sustainability Coach) is in the Presentation Layer - Consumption Management Component, Carbon Calculation Component and Milestone Tracking Component are in the Business Layer - Database Component is in the Data Layer (information management). - Each component is made simple to understand, change and maintain without disturbing other layers.

Reason 2 - Data is well-organized and easier to maintain

The architecture that is presented here flows information through the system in a structured manner. Consumption information sent in from the User Interface is sent to the Consumption Management Component, then to the Carbon Calculation Component, which outputs the results of the CO₂e emission for the Milestone Tracking Component. Business components communicate with the Database Component by means of database queries.

Security Advantage:

Layered Architecture can also address security issues by separating the Presentation Layer from the Data Layer. Since users should be prevented from directly accessing stored data, the access can be enabled via the Business Layer which acts as a middle platform, thereby restricting access to household consumption data, and security controls can be implemented at different levels.

Performance Benefit:

In layered architecture, processing responsibilities are separated at the component level. For example, a Carbon Calculation Component can deal with calculating carbon, while the Database

Component deals with the storing and retrieving of data, avoiding processing between unrelated components, increasing efficiency. As identified on the handout, clear separation of concerns is a benefit of using this style of architecture.